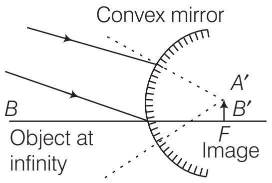
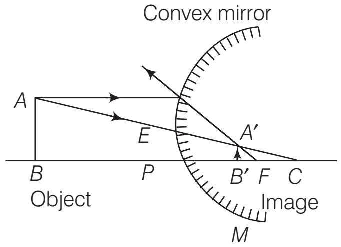
Image Formation by Convex Mirror

(*i*) **When the object is at infinity** The image is formed at focus.



(*ii*) **When the object is between infinity and pole** The image is formed between pole and focus.



MIRROR FORMULA

- If v is the image distance, u is the object distance and f is the focal length of a mirror, then $\frac{1}{v}+\frac{1}{u}=\frac{1}{f}=\frac{2}{R}$
- The linear magnification (m) produced by a mirror is equal to the ratio of the image distance to the object distance with negative sign, $m=-\frac{v}{u}=\frac{f}{f-u}=\frac{f-v}{f}$
- If the magnification has positive sign, then the image is virtual and erect.
- If the magnification has negative sign, then the image is real and inverted.

Uses of Spherical Mirrors

Concave Mirrors

- These are used as reflectors in automobiles (car, buses, etc.), head light, search light, hand torches and table lamps.
- These are used as shaving mirrors.
- These are used by doctors for focussing intense light to examine inside of eye, ears, etc.
- Large concave mirrors are used in the application of solar energy, to focus the sun's rays for heating solar furnaces etc.

Convex Mirrors

- These are used as rear-view mirrors in automobiles as they give large view of the traffic.
- Big convex mirrors are used as shop security mirrors.
- These are used wherever we want to see erect images.

➨ **Note** We cannot use a concave mirror as a rear-view mirror in motor vehicles.

REFRACTION OF LIGHT

When a ray of light passes from one medium to other, it bends from its path. This phenomenon of bending of light is called as refraction of light.

- When light passes from rarer medium to a denser medium, it bends towards the normal at the point of incidence.
- When light passes from a denser medium to a rarer medium, it bends away from the normal.
- When a ray of light enters from one medium to other medium its frequency and phase do not change but wavelength and velocity change.

Laws of Refraction

- The incident ray, the refracted ray and the normal at the point of incidence all lie in the same plane.
- The ratio of the sine of the angle of incidence to the sine of the angle of refraction is a constant for a given medium.

$$\frac{\sin i}{\sin r}=\mu \qquad \text{[Snell's law]}$$

where, μ is refractive index of the second medium with respect to first.

$$\because \quad \mu=\frac{\mu_2}{\mu_1}=\frac{c/v_2}{c/v_1}$$

$$\therefore \quad \mu=\frac{v_1}{v_2} \text{ or } {}_1\mu_2=\frac{v_1}{v_2}$$

where, v_1 is speed of light in first medium and v_2 is speed of light in second medium.

Some Illustrations of Refraction

- Twinkling of stars.
- Oval shape of sun in the morning and evening.
- Bending of a linear object when it is partially dipped in a liquid inclined to the surface of the liquid.
- A fish in a pond when viewed from air appears to be at a smaller depth than actual depth.
- A coin at the base of a vessel filled with water appears raised.

TOTAL INTERNAL REFLECTION

In case of propagation of light from denser to rarer medium through a plane boundary. **Critical angle** is the angle of incidence for which angle of refraction is 90°.

- If light is propagating from denser medium towards the rarer medium and angle of incidence is more than critical angle, then the light incident on the boundary is reflected back in the denser medium, obeying the law of reflection. This phenomenon is called total internal reflection.

Conditions of Total Internal Reflection

- Light must be propagating from denser to rarer medium.
- Angle of incidence must exceeds the critical angle.

• Sparkling of diamond, mirage and looming, shinning of air bubble in water and optical fibre are the examples of total internal reflection.

➨ **Note** Optical fibre is used for the transmitting and receiving the electrical signals after converting it into the light signal.

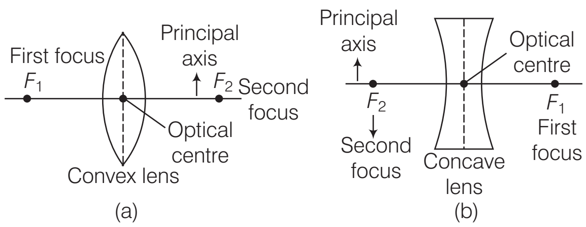
SPHERICAL LENSES

A lens is a piece of transparent glass bounded by two spherical surfaces.

There are two types of lenses

(*i*) **Convex lens (converging lens)** A lens which is thicker at the centre and thinner at its ends.

(*ii*) **Concave lens (diverging lens)** A lens which is thinner at the centre and thicker at its ends.



• The centre point of a lens is known as its optical centre.

• The line joining the centres of the surfaces of the lens is called the principal axis of the lens.

• The distance of the first focus from the optical centre of the lens is called the first focal length of the lens.

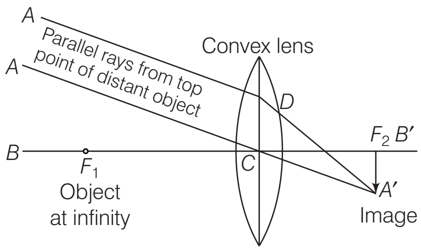
• The distance of the second focus from the optical centre of the lens is called the second focal length or the principal focal length.

Position of object	Position of image	Size of image	Nature of image
At infinity	At F_2	Highly diminished	Real and inverted
Beyond $2F_1$	Between F_2 and $2F_2$	Diminished	-do-
At $2F_1$	At $2F_2$	Same size	-do-
Between $2F_1$ and F_1	Beyond $2F_2$	Enlarged	-do-
At F_1	At infinity	Highly enlarged	-do-
Between F_1 and lens	Behind the object, on the same side of the object	Enlarged	Virtual and erect

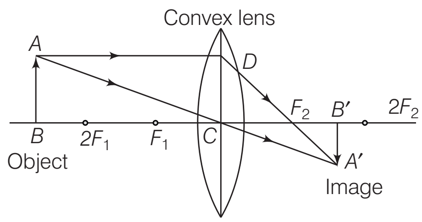
Position of object	Position of image	Size of image	Nature of image
At infinity	At F_2	Diminished	Erect and virtual
Between infinity and lens	Between F_2 and lens	Diminished	Erect and virtual

Image Formation by a Convex Lens

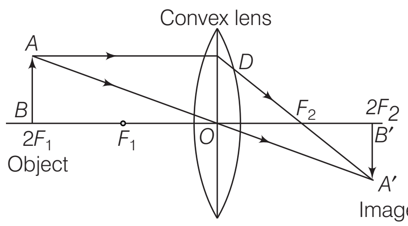
(*i*) **Object** at **infinity** The image formed is at second focus, real, inverted and highly diminished.



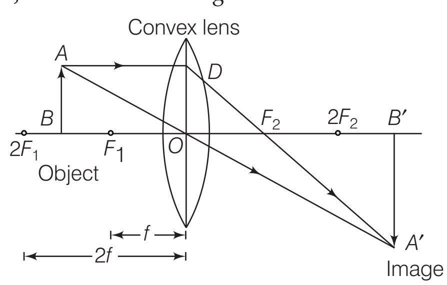
(*ii*) **Object beyond** $2F_1$ The image formed is between F_2 and $2F_2$ on the right side of the lens, real, inverted and diminished.



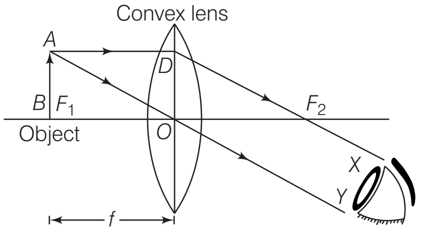
(*iii*) Object at $2F_1$ The image formed is at $2F_2$, real, inverted and of same size as that of the object.



(*iv*) Object between F_1 **and** $2F_1$ The image formed is beyond $2F_2$, real, inverted and magnified.



(*v*) **Object at F_1** The image formed is real, inverted, and highly enlarged at infinity.



(*vi*) **Object between F_1 and O** The image formed is behind the object, virtual, erect and enlarged.

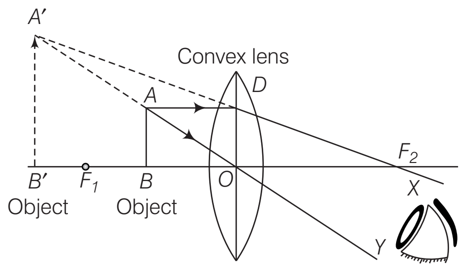
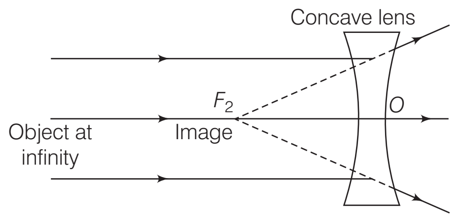
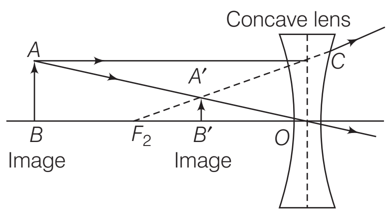


Image Formation by a Concave Lens

(*i*) **Object at infinity** The image formed is at F_2, erect, virtual and diminished.



(*ii*) **Object anywhere between infinity and lens** The image formed is between F_2 and lens, erect, virtual and diminished.



LENS FORMULA

- If v is the image distance, u is the object distance and f is the focal length of a lens, then

$$\frac{1}{f} = \frac{1}{v} - \frac{1}{u}$$

Magnification, $$m = \frac{v}{u} = \frac{I}{O} = \frac{\textit{Size of image}}{\textit{Size of object}}$$

- If the magnification m has a positive value, the image is virtual and erect. And if the magnification m has a negative value the image will be real and inverted.
- If m is the total magnification of two lenses in contact having magnification m_1 and m_2, then

$$m = m_1 \times m_2$$

- Power (P) of a lens is the reciprocal of focal length of lens i.e. $$P = \frac{1}{f}$$

SI unit of power is dioptre (D).

- If two or more lenses are combined, then
  - there is an increase in the magnification of image.
  - the final image is erect.
  - the spherical aberration is reduced.
- If P is power of two lenses in contact having powers P_1 and P_2, then

$$P = P_1 + P_2 \quad \text{or} \quad \frac{1}{F} = \frac{1}{f_1} + \frac{1}{f_2}$$

- If two lenses of equal focal length but of opposite nature are in contact, then combination will behaves as a plane glass plate and $f = \infty$

➡ **Note** When lens is dipped in a liquid of higher refractive index the focal length increases and convex lens behaves as concave lens and *vice-versa*.
An air bubble trapped in water or glass appears as convex but behaves as concave lens.

Properties of Light

Dispersion of Light

Sir Isaac Newton observed that when a narrow beam of light is incident on a prism, the emergent beam is not only deviated but at the same time splits up into a coloured band of seven colours. This phenomenon is called dispersion of light.

- The seven colours of band are Violet, Indigo, Blue, Green, Yellow, Orange and Red (can be memorised as VIBGYOR).
- The colour of an object is determined by the colour of the light reflected by it.
- Violet colour deviates through maximum angle and red colour deviates through the minimum angle.
- Rainbow is formed due to the dispersion of light suffering refractions and total internal reflection in the droplets present in the atmosphere.

Primary Colours

Red, green and blue are called primary colours or basic colours.

Mixing of Colours

Red + Green + Blue = White
Red + Blue = Magenta
Blue + Green = Peacock blue (or Cyan)
Red + Green = Yellow

- Magenta, Cyan and Yellow, etc., are called secondary colours.
- A rose appears red, when day light falls on it because it absorbs all the constituent colour of white light except red which it reflects to us.
- If all the colours of white light are reflected back from the object, then it appears white.
- If all the colours of white light are absorbed by the object, then it appears black.
- The natural colours of transparent objects are due to their selective transmission of light.

Scattering of Light

- When sunlight passes through the earth's atmosphere, much of the light is absorbed by the fine dust particles and air molecules in the atmosphere which give out the absorbed light in some other direction. This is scattering of light.
- The scattering of light by particle in its path is called Tyndall effect.
- The blue coloured light present in white sunlight is scattered much more easily by air molecules causes the blue colour of the sky.
- The sun and the surrounding sky appear red at sunrise and at sunset because at that time most of blue colour present in sunlight has been scattered out and away from our line of sight, leaving behind mainly red colour in the direct sunlight beam that reaches our eyes.
- Dangers signals are of red colour because red colour of light scatter least and therefore signals can be see from far.
- If the earth had no atmosphere, there would have been no scattering of sunlight at all. In that case, no light from the sky would have entered our eyes and the sky would have looked dark and black to us.

Interference of Light

When two waves of exactly same frequency travels in a medium, in the same direction simultaneously, then due to their superposition at same points, intensity of light is maximum while at some other points intensity is minimum. This phenomenon is called interference of light.

- Thin layer of oil on water surface and soap bubbles show various colours in white light due to interference of waves reflected from the two surfaces of the film.
- Polarisation is the only phenomenon which proves that light is a transverse wave.

Diffraction of Light

If an opaque obstacle is kept between a source of light and a screen, a sufficiently distinct shadow is obtained on the screen. This shows that light which travels the obstacle is small. There is a departure from straight line propagation and so the light bends round the corners of obstacle and enters in the geometrical shadow. This bending of the light is called diffraction.

HUMAN EYE

Human eye is an optical instrument which forms real image of the objects.

- The ability of an eye to focus the distant objects as well as the nearby objects on the retina by changing the focal length of its lens is called accommodation.
- Least distance of distinct vision is 25 cm.
- Far point of normal eye is at infinity.

Defect of Human Eye

The ability to see is called vision. The defects of vision are known as defects of eye. There are following defects of human eye

- **Myopia** A person suffering from myopia can see the near objects clearly while far objects are not clear; concave lens is used to remove it. The defect of eye called myopia is caused
  (*i*) due to high converging power of eye lens or
  (*ii*) due to eye ball being too long.
- **Hypermetropia** A person suffering from hypermetropia can see the distant objects clearly but not the near objects. Convex lens is used to remove it. The defect of eye called hypermetropia is caused
  (*i*) due to low converging power of eye lens or
  (*ii*) due to eye ball being too short.
- **Presbyopia** In the old age, the power of accomodation of the eye is so much lost that a person can neither see the near objects distinctly nor the distant objects. To correct this defect bifocal lens is used.
- **Astigmatism** In this defect, a person cannot distantly see the horizontal and vertical lines, simultaneously at a normal distance. To correct this defect, cylindrical lens is used.
- **Cataract** In this defect, an opaque, white membrane is developed on cornea due to which a person losses power of vision partially or completely. This defect can be removed by removing this membrane through surgery.

➨ **Note** Colour blindness is a genetic disorder which occurs by inheritance. John Dalton was colour blind.

Optical Magnifying Instruments Microscope

A microscope is an instrument used to see objects that are too small for the naked eye.

- A simple microscope consists of a convex lens of small focal length.
- It is used to see the details of very small objects.
- The magnifying power of a simple microscope is given by $M = \left[1 + \frac{D}{f}\right]$, D is least distance of distinct vision.
- For compound microscope $m = m_o \times m_e$, where m_o is magnification of objective lens and m_e is magnification of eye-piece.

Telescope

- Telescope is an optical instrument which is used for observing distinct images of heavenly bodies like stars, planets, etc., when the final image is formed at infinity.
- There are two types of telescopes
  - Astronomical telescope (invented by Kepler in 1611)
  - Galilean telescope (invented by Galileo in 1609)
- In an astronomical telescope, the objective lens is a convex lens of large focal length but eye-piece is a convex lens of short focal length.
- In Galilean telescope, the objective lens is a convex lens of large focal length but the eye-piece is a concave lens of short focal length.
- Magnifying power of telescope, $m = \frac{f_o}{f_e}$ where f_o is focal length of objective of the telescope and where f_e is focal length of eye-piece of the telescope.
- Length of telescope tube in astronomical telescope is

$$L = f_o + f_e$$

In Galilean telescope the length of telescope,

$$L = f_o + f_e.$$

➠ **Note** Periscope consists of two plane mirrors inclined at an angle of 45°. The principle of working of periscope is based upon reflection and refraction.

ELECTRIC CURRENT

ELECTRIC CHARGE

Charge is the basic property associated with matter due to which it produces and experience electrical and magnetic effects. The SI unit of electric charge is Coulomb.

- Positively charged particles called protons and negatively charged particles called electrons. A proton possesses a positive charge of 1.6×10^{-19}C, whereas, an electron possesses a negative charge of 1.6×10^{-19}C.
- Similar charges repel each other and opposite charges attract each other.

Conservation of Charges

When two or more charged bodies come in contact, then total charge on all the bodies are conserved.

- **Conductors** are those substances which allow passage of electrical charges to flow through them and have very low electrical resistance e.g. copper, aluminium, gold, silver, etc.
- Human body and earth act like a conductor. Silver is the best conductor.
- Charges are always distributed on the surface of the conductor.
- **Resistors** offer high resistance to the flow of current through them e.g. eureka, nichrome, etc.
- **Insulators** have infinite resistance and do not allow the passage of current. e.g. rubber, glass, ebonite, etc.

➠ **Note** The presence of free electrons in a substance makes it a conductor.

Coulomb's Law

- The force of attraction or the force of repulsion acting between the two point charges is proportional to the product of the magnitudes of the two charges and inversely proportional to the square of the distance between them i.e. $F = \frac{1}{4\pi\varepsilon_0} \frac{q_1 q_2}{r^2}$

Electric Field

The region around an electric charge in which its effect can be experienced is called the electric field.

$$\text{Electric field intensity } (E) = \frac{F}{q_0}$$

Electric Potential

The electric potential at a point in an electric field is the work done in bringing a unit positive charge from infinity to that point.

- Electric potential $= \frac{\text{Work done}}{\text{Charge}}$. Its SI unit is volt (V).
- Potential difference $(V_A - V_B)$ between two points A and B is the work done in bringing a unit charge from point B to point A.
- Potential difference is a scalar quantity and is measured by means of voltmeter (a high resistance device).
- The electric potential inside a spherical surface is same at each point and is equal to the potential on the surface.
- Electrical potential on earth is considered to be zero.
- A voltmeter connected in parallel with conductor to measure the potential difference across its ends.
- Voltage is the other name for potential difference.

Electric Current

- The rate of flow of electric charges through any cross-section of conductor is called electric current.
- The direction of positive charges is same as direction of conventional current.

$$\text{Current} = \frac{\text{Charge}}{\text{Time}} \Rightarrow I = \frac{Q}{t}$$

The SI unit of electric current is ampere.

- Current is measured by an instrument called ammeter.
- An ammeter is connected in series with the conductor to measure the current passing through it.